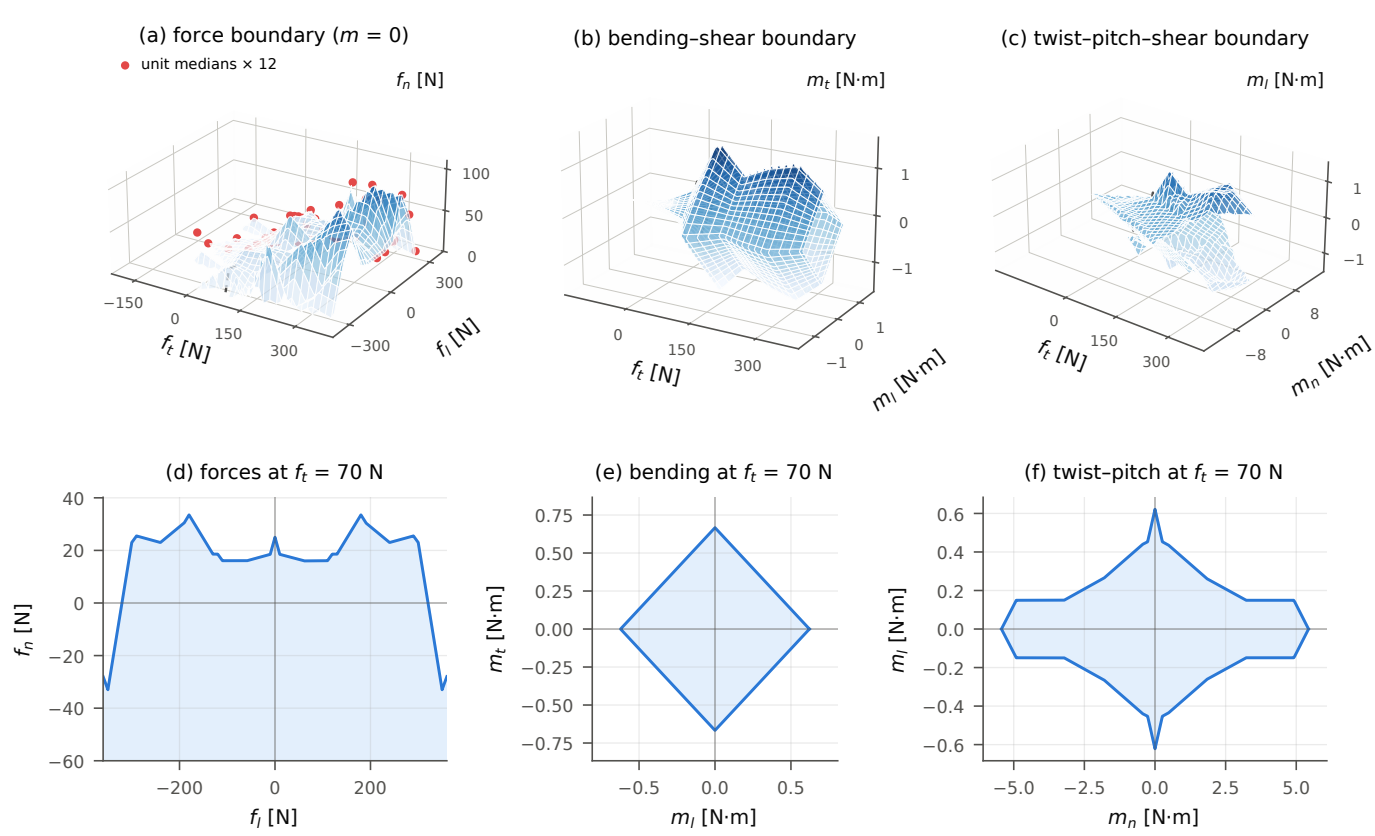

(a) force boundary (m = 0)
unit medians × 12
f_n [N]
100
50
0
−150
0
150
300
f_t [N]
300
0
−300
f_l [N]
(b) bending–shear boundary
m_t [N·m]
1
0
−1
0
150
300
f_t [N]
1
0
−1
m_l [N·m]
(c) twist–pitch–shear boundary
m_l [N·m]
1
0
−1
0
150
300
f_t [N]
8
0
−8
m_n [N·m]
(d) forces at f_t = 70 N
40
20
0
−20
−40
−60
f_n [N]
−200
0
200
f_l [N]
(e) bending at f_t = 70 N
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
m_t [N·m]
−0.5
0.0
0.5
m_l [N·m]
(f) twist–pitch at f_t = 70 N
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
m_l [N·m]
−5.0
−2.5
0.0
2.5
5.0
m_n [N·m]